Kompetenzen

PROGRAMMING

FULL STACK

BACKEND

GENERATIVE AI

DEPLOYMENT

CAREER

KOMPETENZEN TECHNOLOGIES

Python Full Stack Developer with Generative AI

115-Day Project-Driven Career Program

230 Technical Hours + 20 Hours Career Readiness

115 DAYS **230** TECHNICAL HOURS **02** MAJOR CAPSTONES

Build. Integrate. Deploy.

Kompetenzen Technologies

PROGRAMME OVERVIEW

Programme at a glance

A project-driven developer programme designed for students with basic programming knowledge who want to build real full-stack applications and develop the skills to integrate Generative AI into modern software applications.

115	230	20	Project-driven	02
DAYS	TECHNICAL HOURS	HOURS CAREER READINESS	LEARNING MODEL	MAJOR CAPSTONE PROJECTS

~30% CONCEPTS

~70% HANDS-ON PRACTICAL DEVELOPMENT

WHAT THIS PROGRAMME SETS OUT TO DO

01 Learn to build.

02 Learn to integrate AI.

03 Learn to deploy.

04 Build a professional portfolio.

05 Prepare for the AI-enabled developer job market.

PREREQUISITE

Basic programming knowledge and genuine interest in coding/software development.

YOU SHOULD ALREADY UNDERSTAND

Variables, conditions, loops, functions and basic programming logic.

THE IDEAL LEARNER

Who is this programme for?

This programme is suitable for students who:

- Already know basic programming
- Enjoy coding
- Want to become professional programmers/software developers
- Want to build real applications rather than only study theory
- Want to understand backend development
- Want to learn how Generative AI is integrated into software
- Want to develop an AI-enabled developer skill set
- Want a project portfolio
- Want career preparation alongside technical training

PLEASE NOTE

This is not a beginner computer course. It is designed for students who already have a basic programming foundation and want to become developers.

THE LEARNING LOGIC

Why Python + Full Stack + Generative AI?

01 PROGRAMMING FOUNDATION

02 FULL STACK DEVELOPMENT

03 BACKEND ENGINEERING

04 GENERATIVE AI

05 AI APPLICATION DEVELOPMENT

06 DEPLOYMENT

07 CAREER READINESS

Modern developers increasingly work with AI-assisted development and AI-powered application features. This programme does not stop at using AI tools. You first learn how software is built — and then how to integrate AI into real applications.

The goal is not simply to use AI tools.
AI-assisted development, not AI-dependent development.

WHAT YOU WILL WORK WITH

Technology stack

FRONTEND

HTML5 · CSS3 · Bootstrap · Vanilla JavaScript

BACKEND

Python · Django · Django REST Framework

DATABASE

MySQL

DEVELOPMENT TOOLS

Git · GitHub · Postman · MySQL Workbench or equivalent

GENERATIVE AI

LLM APIs · Prompt Engineering · Structured Outputs · Embeddings · Vector Search · FAISS · ChromaDB · RAG · Tool / Function Calling · AI Agents · LangChain / AI Application Frameworks · AI Evaluation · AI Security

ABOUT REACT

React is not part of the core 115-day curriculum.

The core frontend approach is **HTML + CSS + Bootstrap + Vanilla JavaScript**.

React may be available separately as an optional recorded/self-study module.

10 PHASES · 115 DAYS

Your programme journey



01	WEB & FRONTEND FOUNDATION	Days 1–10	20 Hrs
02	PYTHON PROGRAMMING & OOP	Days 11–32	44 Hrs
03	SQL & DATABASE ENGINEERING	Days 33–40	16 Hrs
04	DJANGO FULL-STACK DEVELOPMENT	Days 41–58	36 Hrs
05	REST API, JWT & RBAC	Days 59–68	20 Hrs
06	GENERATIVE AI FOUNDATIONS	Days 69–76	16 Hrs
07	LLM APPLICATION ENGINEERING	Days 77–86	20 Hrs
08	EMBEDDINGS, RAG & KNOWLEDGE SYSTEMS	Days 87–98	24 Hrs
09	TOOLS, AGENTS, SECURITY & PRODUCTION AI	Days 99–108	20 Hrs
10	FINAL CAPSTONE, DEPLOYMENT & PORTFOLIO	Days 109–115	14 Hrs

TOTAL **115 Days** **230 Technical Hours** + 20 Hours Career Readiness

Highlighted phases end with a major capstone project.

PHASE 01

01 Web & Frontend Foundation

Days 1–10 | **20** HOURS

01	02	03	04	05	06	07	08	09	10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will learn how web pages are structured, styled and made interactive — the frontend foundation for understanding how a full-stack application talks to its backend.

KEY TOPICS

HTML5 · Semantic HTML · Forms · Accessibility basics · CSS · Box model · Flexbox · Responsive design · Bootstrap · JavaScript fundamentals · DOM · Events · Fetch API basics · JSON · Browser storage

PRACTICAL OUTCOMES

- **01** Static website
- **02** Responsive Bootstrap interface
- **03** Interactive JavaScript application
- **04** API-connected browser application

PHASE 02

02 Python Programming & OOP

Days 11–32 | **44** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will develop strong programming logic and solid Python development capability, from fundamentals through object-oriented programming.

KEY TOPICS

Python fundamentals · Variables · Data types · Operators · Conditions · Loops · Functions · Lists · Tuples · Sets · Dictionaries · Comprehensions · Problem solving · Debugging · File handling · CSV · JSON · Exception handling · Modules · Packages · Object-Oriented Programming · Classes · Objects · Encapsulation · Inheritance · Polymorphism · Abstraction · Composition

PRACTICAL OUTCOME

01 Completed Python application

PHASE 03

03 SQL & Database Engineering

Days 33–40 | **16** HOURS

01	02	03	04	05	06	07	08	09	10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will learn to design relational databases, write SQL and connect databases to Python applications.

KEY TOPICS

Relational databases · MySQL · Tables · Primary keys · Foreign keys · Constraints · Schema design · SQL CRUD · DDL · DML · SELECT · WHERE · ORDER BY · GROUP BY · JOIN · Relationships · Normalization · Python + MySQL integration · ORM concepts

PRACTICAL OUTCOME

01 Database-backed Python application

PHASE 04

04 Django Full-Stack Development

Days 41–58 **36** HOURS

01 02 03 **04** 05 06 07 08 09 10

Day 1 | Day 68 · Milestone 1 Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will learn to build server-side web applications with Django — from architecture and data models to forms, CRUD and authentication.

KEY TOPICS

Django architecture · Project structure · Applications · MVT · URLs · Views · Templates · Models · Migrations · QuerySets · ORM · Relationships · Forms · ModelForms · Validation · CRUD · Template inheritance · Authentication · Sessions · Users · Permissions

PRACTICAL OUTCOME

01 Student/Course Management System

PHASE 05

05 REST API, JWT & RBAC

Days 59–68 | **20** HOURS

01 | 02 | 03 | 04 | **05** | 06 | 07 | 08 | 09 | 10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will learn to build professional REST APIs and secure them with token-based authentication and role-based access.

KEY TOPICS

- REST concepts
- Django REST Framework
- Serializers
- API views
- ViewSets
- Routers
- Filtering
- Search
- Pagination
- JWT authentication
- Protected endpoints
- Role-based access control
- Student/Trainer/Admin roles
- Permissions

CAPSTONE 1 TECHNOLOGY

- HTML
- Bootstrap
- Vanilla JavaScript
- Python
- Django
- DRF
- MySQL
- JWT
- RBAC

MAJOR PRACTICAL OUTCOME

01 Capstone 1 — Full-Stack Management System

PHASE 06

06 Generative AI Foundations

Days 69–76 | **16** HOURS

01 | 02 | 03 | 04 | 05 | **06** | 07 | 08 | 09 | 10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will gain a practical, developer-level understanding of Generative AI and how AI applications are put together.

KEY TOPICS

AI | Machine Learning | Deep Learning | Generative AI | Generative models | Large Language Models | Tokens | Context window | Inference | Parameters | Transformer concepts | Attention concepts | LLM capabilities | Hallucination | Model limitations | Model selection | Generative AI application architecture

YOU WILL UNDERSTAND THE RELATIONSHIP

LLM + PROMPT + CONTEXT + TOOLS + RETRIEVAL + APPLICATION

PRACTICAL OUTCOME

01 Basic AI chatbot prototype

PHASE 07

07 LLM Application Engineering

Days 77–86 | **20** HOURS

01 | 02 | 03 | 04 | 05 | 06 | **07** | 08 | 09 | 10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will learn to program applications using modern LLM APIs — including prompt engineering, structured outputs and AI-assisted development.

KEY TOPICS

- LLM APIs
- API authentication
- API requests/responses
- Model selection
- Error handling
- Rate limits
- Prompt engineering
- Zero-shot prompting
- Few-shot prompting
- Role/context/instructions
- Prompt templates
- Prompt chaining
- Structured outputs
- JSON
- Schemas
- Parsing
- Validation
- AI-assisted coding
- AI-assisted debugging
- AI-assisted refactoring
- AI-assisted testing
- AI-assisted documentation
- AI-assisted research

Key philosophy

AI-assisted development, not AI-dependent development.

PRACTICAL OUTCOME

01 AI Assistant or Resume Data Extractor

PHASE 08

08 Embeddings, RAG & Knowledge Systems

Days 87–98 | **24** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | **08** | 09 | 10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will learn to build AI systems that retrieve and use external, domain-specific knowledge through embeddings, vector databases and RAG.

KEY TOPICS

Embeddings · Vector representations · Semantic similarity · Similarity search · Vector databases · FAISS · ChromaDB · Indexing · Metadata · Document ingestion · Text extraction · Cleaning · Chunking · RAG · Retrieval · Context construction · Generation · Grounding · Advanced RAG concepts · Retrieval quality · RAG evaluation · Relevance · Groundedness · Hallucination

PRACTICAL OUTCOMES

01 Semantic search prototype

02 Vector search application

03 Document ingestion pipeline

04 Working RAG chatbot

PHASE 09

09 Tools, Agents, Security & Production AI

Days 99–108 | **20** HOURS

01 02 03 04 05 06 07 08 **09** 10

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will explore advanced AI application engineering — tools, agents, multimodal AI, architecture decisions, security and production considerations.

KEY TOPICS

Tool/function calling · Tool selection · Function execution · AI agents · Planning · Multi-step tasks · AI workflows · LangChain · Retrievers · Agents · Multimodal AI · Image understanding · Document understanding · Vision-language concepts · Audio/speech concepts · Prompting vs RAG vs tools vs agents · Fine-tuning concepts · AI security · API key security · Prompt injection · Data leakage · Input/output validation · User isolation · Deployment concepts · Logging · Token usage · Cost · Latency · Reliability

PRACTICAL OUTCOMES

01 Tool-using assistant

02 Simple AI agent workflow

03 Multimodal AI mini-demo

04 AI architecture decision exercise

05 Secured AI workflow

06 Production-readiness checklist

PHASE 10

10 Final Capstone, Deployment & Portfolio

Days 109–115 | **14** HOURS

01 02 03 04 05 06 07 08 09 **10**

Day 1 | Day 68 · Milestone 1 | Day 115 · Capstone 2

WHAT YOU WILL LEARN

You will build, secure, evaluate, deploy and present your final AI application — the AI Document Assistant.

KEY TOPICS

Django · Django REST Framework · LLM API · Embeddings · Vector database · RAG · Authentication · Per-user document ownership · Chat functionality · Security · Evaluation · Deployment · Frontend: Bootstrap + Vanilla JavaScript

FINAL OUTCOMES

01 Working AI application

02 Deployed application

03 GitHub repository

04 Project documentation

05 Architecture explanation

06 Portfolio-ready project

07 Final technical presentation

DEMONSTRABLE PROJECT WORK

Two major capstone projects

Capstone 1

END OF PHASE 05 · DAY 68

Full-Stack Management System

YOU DEMONSTRATE

Frontend · Backend · Database · CRUD · REST API · Authentication · JWT · RBAC

Capstone 2

PHASE 10 · DAY 115

AI Document Assistant

YOU DEMONSTRATE

Document ingestion · Chunking · Embeddings · Vector database · Retrieval · RAG · LLM integration · User-specific document access · Chat · Security · Evaluation · Deployment

Students do not finish with only certificates.

They finish with demonstrable project work.

IN ADDITION TO 230 TECHNICAL HOURS

Career readiness — 20 hours

The objective is to help you communicate your technical knowledge, present your projects, prepare a professional profile and become interview-ready.

NO.	MODULE	HOURS
01	Professional Communication	4 Hours
02	Spoken English / Workplace Language	4 Hours
03	Technical Communication & Presentation	3 Hours
04	Resume & ATS Readiness	3 Hours
05	Interview Preparation	4 Hours
06	Workplace Behaviour & Career Planning	2 Hours
Total career readiness		**20 Hours**



Technical training

Career readiness

PLACEMENT SUPPORT AND CAREER ASSISTANCE

Kompetenzen digital career & placement ecosystem

As a student, you can access the Kompetenzen digital placement ecosystem for:

01

Job discovery

Explore job opportunities listed on the Kompetenzen platform.

02

Direct application

Apply directly to relevant opportunities.

03

ATS resume

Create and maintain an ATS-friendly resume.

04

Career profile

Maintain a professional profile.

05

Consultant support

Connect with Kompetenzen career/placement consultants for guidance and support.

06

Career readiness

Use the Career Readiness programme to prepare for applications and interviews.

These are placement support and career assistance services. They do not constitute a guarantee of employment.

YOUR PORTFOLIO PROGRESSION

What you will be able to build

01 FRONTEND APPLICATION → 02 PYTHON APPLICATION → 03 DATABASE APPLICATION

04 DJANGO APPLICATION → 05 REST API APPLICATION → 06 AI ASSISTANT

07 SEMANTIC SEARCH APPLICATION → 08 RAG APPLICATION → 09 AI AGENT WORKFLOW

10 DEPLOYED AI DOCUMENT ASSISTANT

FINAL STUDENT OUTCOME

> By the end of the programme, students will have developed the skills to build full-stack applications using Python and Django, work with databases and REST APIs, implement authentication and role-based access, integrate LLMs, build RAG-based applications, work with AI tools and workflows, apply AI security and evaluation principles, deploy an AI application and present their work professionally.

Build. **Integrate.** **Deploy.** **Present.**